

Shaibal Muhtadee

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EDUCATION

University of Toronto

Bachelor of Applied Science in Computer Engineering

Toronto, ON

Sept 2020 – Apr 2025

EXPERIENCE

Software Engineering Intern

Zebra Technologies

May 2023 – Sept 2024

Toronto, ON

- Built authentication and authorization services in Rust to secure API access for over 200 autonomous robots
- Developed TypeScript APIs for securely storing and retrieving API keys, JWTs, and robot configuration data
- Implemented attribute-based access control with a custom policy parser for authorization rule updates
- Created React and Flutter interfaces for managing secrets, access policies, and robot configurations
- Built a CLI that retrieved and connected secrets from GitHub, GCP, and AWS for development workflows
- Wrote Rust integration tests with 99% code coverage for auth flows, APIs, and service communication
- Containerized Rust services with Docker and maintained CI/CD pipelines for Kubernetes deployments

PROJECTS

Inokta | *Independent Startup Project, Next.js, FastAPI, TypeScript, Python*

June 2025 – June 2026

- Co-developed an AI workspace that turns documents into an editable world model of entities and relationships
- Created Alice, a LangGraph AI harness that extracts entities and generates persistent text, tables, and graphs
- Developed document pipelines for PDF extraction, table detection, chunking, embeddings and ChromaDB retrieval
- Implemented an interface for editing entities and relationships and correcting AI-inferred connections
- Streamed AI responses and tool activity over WebSockets with session-scoped conversations and MCP tools

GanttWise | *Python, React, Node.js, MongoDB*

Sept 2024 – Apr 2025

- Built dynamic web app using React and TypeScript to automate task planning with Deep Reinforcement Learning
- Used Python scripts to preprocess datasets and debug learning agents, optimizing scheduling under constraints
- Integrated a full-stack pipeline with Node.js backend and MongoDB, enabling real-time project updates

Searchington | *Python, Flask, Redis, AWS*

Sept 2024 – Dec 2024

- Built a web search engine with a crawler that recursively visits URLs, extracts HTML, creates word lists, and stores data in SQLite, using an inverted index and various metrics to deliver relevant search results
- Hosted on AWS and implemented caching with Redis to optimize query performance, improving response times by over 70%

ChromaMap | *C++, OpenStreetMap API, GTK Library, Git*

Sept 2022 – Apr 2023

- Created a Geographic Information System mapping software using C++ and the OpenStreetMap API
- Implemented Dijkstra algorithm and A* pathfinding heuristics to solve the travelling salesman problem
- Improved browse and load times by over 300% by developing a Level of Detail tiling scheme for the map
- Optimized application speed by multi-threading and code visualization, stack trace and flame graphs

TECHNICAL SKILLS

Languages: TypeScript, Python, Rust, C++, SQL, HTML/CSS, Bash

Frameworks: React, Next.js, FastAPI, Node.js, Flask, Flutter, LangGraph

Developer Tools: Docker, Kubernetes, AWS, GCP, Linux, Git, CI/CD

Libraries: MongoDB, Redis, ChromaDB, SQLite, REST APIs, WebSockets